C-Suite

Can a board of AI agents produce a fruitful fiscal year?

...kinda?

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Abstract

We replace a Fortune 500 company's entire C-suite with large language model agents and run them through a fiscal year.

Ten AI Executive (AIE) agents — each assigned a distinct C-suite role, behavioral archetype, and priority weight vector — propose strategic actions, vote on them using a weighted scoring system, and update a simulated company state each quarter. The simulation seeds from real FY22Q4 financial data and runs four quarters with no real FY2023 data after initialization.

The AI board correctly identified the right strategic direction in every run, independently converging on AI-heavy, cloud-forward investment without being told that is what the real company did. It then proceeded to overspend at a rate that would concern any real CFO. To be fair, this is partly my fault: certain financial mechanisms (share buybacks, debt management, dividend practices) were not included in the simulation's action space, either because the data is not publicly available at the granularity needed, or because modeling a full capital allocation strategy was firmly out of scope for one person with a Claude API key (and $10 of credits). The gap between simulated and actual cash reserves is best understood as a known design limitation rather than evidence that AI executives cannot count.

Two ablation studies tested a conservative CFO configuration and a CFO/CTO personality swap. The personality swap produced the most interesting result: emergent self-correction in Q4, where the board reversed course without external instruction after accumulating enough bad state to notice.

Strategy, it turns out, is the easy part. Anyone can have a vision. Knowing when to stop spending is a different skill entirely.

1. Introduction

The technology industry has spent considerable energy discussing whether artificial intelligence will replace knowledge workers. Curiously, this conversation has largely spared the people having it. C-suite executives — the highest-paid, most consequential, and most conference-keynote-delivering decision-makers in large organizations — have remained conspicuously absent from automation discourse.

This paper asks the obvious question: what happens if we replace their roles with AI?

The inspiration was a blog post tracking the ratio of a major AI company's hiring activity to its public statements about AI replacing jobs [2]. The frequent claims of SWE job automation expected to come in varying months (some have said it will come in 6, 12, etc.) has been consistently contradicted with the rate at which SWE jobs are being posted.

A multi-agent simulation was constructed in which ten LLM agents govern a major technology company through fiscal year 2023. The simulation seeds from real FY22Q4 financial data. After that, the AI board is on its own. Outcomes are compared against what the actual human executives produced.

The research questions are:

1. Does an AI-simulated C-suite produce decisions measurably different from its human-led counterpart?
2. Do those decisions lead to better, worse, or statistically indistinguishable simulated outcomes?

A baseline run and two ablation studies are reported. The short answer is: the AI board got the strategy right and then could not stop itself from spending. Whether this disqualifies it from executive leadership or uniquely qualifies it is left as an exercise for the reader. We'll also dig into some reasons as to why spending was an issue in the simulation (hint: building a full-fledged simulation of a real fiscal year is hard!).

2. Related Work

This work sits at the intersection of multi-agent LLM systems, corporate strategy simulation, and the genre of research that asks uncomfortable questions with a straight face.

Role-playing and persona assignment in LLMs have been shown to produce meaningfully differentiated behavior across agents, consistent with sociological role theory as articulated by Ralph Linton [1] — the observation that behavior is context-specific and shaped by socially defined categories. Each agent gets a title, an archetype, and a weight vector, and is expected to behave accordingly.

The author is not aware of prior work that has attempted to run a full AI C-suite through a real company's fiscal year and compare outcomes to actuals. This is assumed to be because it seemed like a bad idea. It was done anyway.

3. System Design

3.1 Agent Design

Ten AIE agents map to real C-suite titles from the simulated company's actual leadership structure. The company had 14 additional leadership roles, which were omitted on the grounds that ten LLM agents simultaneously proposing actions is already quite a lot.

Each agent is defined by three properties: a role archetype (prose description of behavioral personality injected into the proposal prompt), a primary domain (the decision category in which the agent holds domain expertise), and an AES weight vector (seven signed integers in [-4, +4] encoding role priorities).

The archetypes are professionally recognizable stereotypes: the CFO is risk-sensitive and financially conservative; the CTO is technically ambitious and AI-forward; the CMO is growth- and brand-oriented. These are abstractions used to operationalize role-differentiated decision-making, not empirical claims about any individual.

Role	Primary Domain	Rev	Op Cost	Cash	HC	R&D	Brand	Innov
CEO	Revenue	4	-1	2	1	3	3	3
CFO	Cash Reserves	2	-4	4	-1	-1	1	1
VP & Chair	Brand Strength	2	-1	2	1	1	3	1
COO	Operating Cost	3	-3	2	2	1	1	1
CPO	Headcount	1	-1	1	4	2	2	2
EVP Strategy	Innovation Index	3	-1	2	1	3	2	4
CMO	Brand Strength	3	-1	1	1	1	4	2
CCO	Revenue	4	-1	2	2	1	3	1
CTO	R&D Investment	2	-1	1	2	4	1	4
EVP Global Sales	Revenue	4	-2	2	2	1	3	1

Table 1. AIE agent AES weight vectors. Positive = agent prefers increases; negative = agent prefers decreases. The CFO's -4 on operating cost is doing a lot of work.

3.2 CompanyState Vector

The company's state at any point in the simulation is a structured JSON object with five sub-sections. Each field is sourced one of three ways: **DIRECT** (parsed from real financial filings, exact figures), **CLAUDE-DERIVED** (Claude reads qualitative documents and converts them to quantitative signals on a 1-10 scale), or **MIXED** (some fields in the section are direct, others are Claude-derived).

- **Financials (DIRECT):** revenue, cost_of_revenue, gross_margin, operating_income, net_income, cash_and_equivalents, total_operating_expenses, rd_spending, sales_marketing_spending, capex. Parsed directly from real earnings XLS files. These numbers are real and we did not touch them.
- **Segments (DIRECT):** productivity_revenue, intelligent_cloud_revenue, personal_computing_revenue. Also real.
- **Human Impacts (MIXED):** total_employees is direct, pulled from annual reports. The remaining fields — hiring_freeze, layoffs_this_quarter, engineering_headcount, sales_headcount — are Claude-derived via Prompt C from the earnings transcript. Treat them as informed estimates.
- **Growth Signals (CLAUDE-DERIVED):** ai_investment_focus, innovation_index, competitive_pressure, regulatory_pressure, brand_strength. Each scored 1-10 by Claude. A score of 5 means the documents didn't give Claude enough to work with.
- **Market Signals (CLAUDE-DERIVED + direct):** investor_sentiment, growth_expectation, stock_price. Stock price was not used in final runs — the AI board does not have a compensation package and therefore does not care about it.

The simulation seeds from real FY22Q4 data. After that, no real FY2023 data enters the loop. Each quarter starts from the previous quarter's simulated end state, which means decisions (and mistakes) compound forward.

3.3 Action Library

The action space consists of 30 discrete actions across eight categories. Agents may only propose actions from this list. This constraint exists because testing without it resulted in one agent proposing actions outside the intended scope of a quarterly corporate strategy cycle.

- **R&D Investment**: increase_rd_5, increase_rd_10, launch_major_ai_initiative, acquire_ai_startup, launch_experimental_product, accelerate_product_development
- **Revenue**: expand_enterprise_sales, expand_global_markets, strategic_partnership, shift_budget_to_high_growth_segments
- **Headcount**: freeze_hiring, layoff_5_percent, layoff_10_percent, increase_engineering_hiring, expand_sales_team, restructure_engineering_teams
- **Operating Cost**: reduce_costs_5, reduce_costs_10, optimize_operations, consolidate_product_lines
- **Cash Reserves**: stock_buyback_program, increase_dividend
- Plus Brand Strength, Innovation Index, and Multi-category / Governance actions.

Notably absent: financing decisions. No debt issuance, no capital management actions beyond the two Cash Reserves entries. This omission was not accidental: the data required to model a realistic capital allocation strategy either is not publicly available at quarterly granularity or requires a level of financial modeling that is well outside the scope of this project. The consequences of this omission will be discussed at length in the results section, the discussion section, and the limitations section. Consider this a preview.

4. Decision Pipeline

Each quarter proceeds through four stages. All LLM calls use claude-sonnet-4-20250514. The total API cost for a full four-quarter simulation run is approximately the price of one cup of coffee, which is significantly less than the fully-loaded compensation of the executives being simulated.

4.1 Qualitative Extraction (Prompt C)

Before the simulation begins, Prompt C processes real earnings documents for FY22Q4 (earnings transcript, press release, and product release list) to populate the derived fields of the CompanyState. This is the only time real qualitative data enters the simulation.

The prompt instructs Claude to return structured JSON with a derivation_notes field explaining each value. Where documents provide insufficient signal, the prompt instructs null over a guess. Claude complied with this instruction approximately 80% of the time.

A note on the earnings slides: the company's PPTX files are fully image-based. Every slide is a picture where python-pptx extracted zero text from them! The earnings transcript was used instead and contains all the same information in a more parseable format.

4.2 Proposal Phase (Prompt D)

Each of the ten AIE agents receives a context packet containing the full CompanyState as markdown tables, an external market context string describing the real macro and technology environment for the quarter (ChatGPT's November 2022 launch, generative AI's position on the Gartner Hype Cycle [4], competitive cloud dynamics), and the ActionLibrary. Agents propose up to three actions with rationale.

All ten agents are called sequentially. Each proposes up to three actions grounded in their role priorities and the current CompanyState. The CTO invariably wants to spend more on R&D. The CFO invariably wants to spend less on everything. The CEO wants to do both at scale. The board keeps it moving!

4.3 Effect Prediction (Prompt A)

For each unique proposed action, Prompt A predicts the directional effect on every CompanyState variable, returning integers in [-3, +3]. The prompt requests consideration of second-order effects. Claude reliably identifies that increasing R&D will decrease cash and improve innovation_index over time.

These predictions are the raw material for the AES scoring. They are generated by Claude based on the current company context and general business reasoning. They are not sourced from empirical data.

4.4 Voting: AES and AWS

The Action Evaluation Score (AES) measures how well a proposed action aligns with an individual agent's priorities:

$$AES_{agent}(a) = \sum_{i=1}^{n} w_{agent,i} * Effect_{i}(a)$$

Where i indexes over the seven AES categories, w_agent,i is the agent's role-priority weight for that category (a field within the CompanyState vector, range -4 to +4), and Effect_i(a) is the sum of Claude-predicted effects for all CompanyState variables in that category. The AES is a signed float. Positive means the agent likes it. Negative means the agent does not.

Vote direction is derived from the sign of the AES. The Authority Weight System (AWS) then scales each vote by domain expertise:

`VoteWeight = VoteDirection × (1 + DomainBonus)`

Domain bonus = 2 if the action's primary category matches the agent's primary domain, else 0. A domain-expert YES contributes +3. A domain-expert NO contributes -3. Three agents share the Revenue domain (CEO, CCO, EVP Global Sales), meaning revenue-oriented actions carry structural momentum when this bloc agrees.

An action passes if FinalDecisionScore > 0. The top five passing actions are implemented each quarter.

$$FinalDecisionScore(ProposedAction\ a) = \sum_{agents,i}^{n} VoteWeight_{agent,i}$$

$$\begin{aligned}
&if\ \ FinalDecisionScore(a)\ >\ Threshold:\\
&\quad CurrentVote.pass(ProposedAction\ a)\\
&else:\\
&\quad CurrentVote.veto(ProposedAction\ a)
\end{aligned}$$

4.5 State Transition (Prompt B)

Prompt B receives the current CompanyState, the approved action list, and external context, and returns an updated CompanyState. Financial actions scale proportionally (increase_rd_10 multiplies rd_spending by 1.10). Prompt B is instructed to apply second-order effects where appropriate.

Prompt B does not project revenue forward. This is by design. Revenue is a lagging indicator: it is the cumulative result of decisions made over multiple quarters, customer purchasing cycles, market conditions, and factors entirely outside any single quarter's executive decisions. Having Claude guess a revenue number from one quarter of approved actions would add noise and false precision to a comparison that is already imprecise. The simulation is designed to compare strategic decision-making behavior: what actions were chosen, how investment was allocated, what the board prioritized. It is not a financial forecasting model, and treating it as one would be a category error.

5. Results

Three simulation runs were completed. All seed from the same real FY22Q4 CompanyState and compound forward with no real FY2023 data after initialization. All are single-trial runs.

5.1 Baseline: Standard 10-Agent Board

Quarter	Rev Sim	Rev Actual	Delta	Margin Sim	Margin Actual	Delta
FY23Q1	$53,458M	$52,747M	+$711M	39.6%	38.7%	+1.0%
FY23Q2	$55,531M	$52,857M	+$2,674M	39.7%	42.3%	-2.6%
FY23Q3	$58,808M	$56,189M	+$2,619M	37.3%	43.2%	-5.8%
FY23Q4	$64,688M	$56,189M	+$8,500M	34.5%	43.2%	-8.7%

Table 2. Baseline revenue and operating margin. Q1 delta of +$711M is respectable. Q4 delta of +$8.5B is the board having opinions.

Quarter	Cash Sim	Cash Actual	Delta	HC Sim	HC Actual	Delta
FY23Q1	$12,431M	$15,646M	-$3,215M	232,050	221,000	+11,050
FY23Q2	$11,431M	$26,562M	-$15,131M	235,971	221,000	+14,971
FY23Q3	$9,876M	$34,704M	-$24,828M	251,469	228,000	+23,469
FY23Q4	$8,235M	$34,704M	-$26,469M	264,343	238,000	+26,343

Table 3. Baseline cash and headcount. The board ended Q4 with $8.2B in cash vs the actual $34.7B, and 26,000 more employees than the real company. These figures are related.

Revenue - correct trajectory, progressive overshoot. Q1 delta was +$711M (+1.3%). By Q4 the simulation was $8.5B over actual. The board approved enterprise sales expansion, cloud investment, and AI initiatives every quarter — good ideas, applied with increasing enthusiasm as the quarters progressed.

Operating margin - converged then diverged. Q1 margin was within 1.0% of actual. By Q4 the gap was -8.7%. The board kept approving growth actions without the operational efficiency gains the real company achieved alongside them. Operational efficiency is difficult to encode in a 30-action library.

Cash - the expected structural problem. The simulation depleted cash every quarter, ending at $8.2B vs the actual $34.7B. As described in Section 3.3, the ActionLibrary does not include financing decisions. The real company maintained its cash position through share buybacks, debt management, and dividend practices that were simply not available to the simulated board.

Headcount - overestimated and compounding. The gap grew from +11K in Q1 to +26K by Q4, as the board approved increase_engineering_hiring alongside growth actions every quarter with no mechanism to self-limit.

Strategic direction - the headline result. Across all four quarters, the board consistently approved launch_major_ai_initiative, expand_cloud_investment, increase_rd_10, and increase_capex_datacenters — without access to any FY2023 data. The real company made precisely these investments during the same period. The board found the right answer without being told what it was.

5.2 Ablation Run 1: Conservative CFO

The CFO's AES weights were made significantly more restrictive: headcount weight dropped from -1 to -3, rd_investment from -1 to -3, and revenue from 2 to 1. The role archetype was updated to "extremely risk-averse and cash-protective." All other agents remained unchanged.

Hypothesis: heavier NO votes from the CFO should reduce cash depletion and margin compression.

Quarter	Rev Sim	Rev Actual	Margin Sim	Margin Actual	HC Sim	HC Actual
FY23Q1	$54,358M	$52,747M	39.9%	38.7%	226,100	221,000
FY23Q2	$56,576M	$52,857M	39.6%	42.3%	228,750	221,000
FY23Q3	$60,205M	$56,189M	41.0%	43.2%	235,600	228,000
FY23Q4	$63,215M	$56,189M	38.3%	43.2%	247,380	238,000

Table 4. Conservative CFO results. Q4 cash: $9,420M simulated vs $34,704M actual. Behavioral tuning improved headcount significantly. It did not move cash meaningfully.

Headcount improved substantially: the Q3 gap dropped from +23,469 to +7,600, and Q4 dropped from +26,343 to +9,380. Operating margin held better through Q3 (41.0% vs 35.8% baseline). The CFO's heavier NO votes changed behavior in the areas where behavior was the problem.

Cash, however, was nearly identical to baseline: Q4 ended at $9,420M vs baseline $8,235M, a $1,185M improvement against a $25.3B gap with actual. This confirms what was expected: behavioral tuning cannot fix a structural gap. The conservative CFO had no access to stock_buyback_program as a cash management tool because the board's version of stock_buyback_program does not generate cash inflows. It was defined as a capital return action, not a financing action.

5.3 Ablation Run 2: Personality Swap (CFO and CTO)

The CFO received the CTO's AES weights and role archetype (technically ambitious, AI-forward, high R&D priority). The CTO received the CFO's AES weights and role archetype (risk-sensitive, financially conservative). Titles and primary domains were kept the same.

Hypothesis: this tests whether role identity (the title seen in the proposal prompt) or the AES weight vector drives board behavior. If the swapped CFO pushes R&D and the swapped CTO vetoes spending, weights dominate. If both continue behaving like their titles, role identity dominates.

Quarter	Rev Sim	Rev Actual	Delta	Cash Sim	HC Sim	HC Actual
FY23Q1	$53,758 M	$52,747 M	+$1,011 M	$12,431 M	221,000	221,000
FY23Q2	$56,046 M	$52,857 M	+$3,189 M	$10,856 M	234,000	221,000
FY23Q3	$58,248 M	$56,189 M	+$2,059 M	$9,534 M	245,700	228,000
FY23Q4	$54,308 M	$56,189 M	-$1,881 M	$12,931 M	226,000	238,000

Table 5. Personality swap results. Q4 is the interesting quarter: the only one in any run where cash increased, revenue undershot actual, and the board appeared to notice it had a problem.

Q4 is the finding. After three quarters of overspending, the board entered Q4 with $9,534M in cash. It responded with more conservative actions: headcount was cut by approximately 12,000, cost-discipline measures were approved, and cash recovered to $12,931M, the only quarter in any run where cash increased. Revenue undershot actual for the first time (-$1,881M). Operating margin rebounded from 35.8% to 39.5%.

The board never received a memo saying, “you are running out of cash.” It saw a deteriorating CompanyState and responded differently than it had in prior quarters. This emergent self-correction was not scripted and was not present in the baseline run.

On the role identity vs weights question: both matter, at different stages. The proposal phase appears sensitive to the title. The CFO agents generated conservative-sounding proposal language even with CTO-style weights, suggesting that role identity shapes what gets proposed. The voting phase appears more sensitive to weights. The swapped CFO applied less resistance to R&D actions than the baseline CFO, consistent with its CTO-style weight vector. The two mechanisms modulate different parts of the pipeline independently.

6. Discussion

6.1 What the AI Board Got Right

Strategic direction. The board independently converged on AI-heavy, cloud-forward investment in every run without access to any FY2023 data. The real company made precisely these investments.

Revenue trajectory was also consistently correct. The simulated company grew revenue in the right direction every quarter. Q1 margin in the baseline was within 1.0% of actual. Headcount matched exactly in Q1 across multiple runs. The board was well-calibrated from the seed state and maintained directional correctness even as magnitudes diverged.

The external context injection (ChatGPT's launch, the Gartner Hype Cycle positioning of generative AI) appears to have driven the board toward the correct strategic posture. Whether this constitutes genuine strategic reasoning or very expensive pattern matching is a philosophical question best left to a different venue.

6.2 What the AI Board Got Wrong

Cash depletion was persistent across all runs and configurations. The Q4 gap ranged from $21.8B (personality swap) to $26.5B (baseline).

This deserves an honest accounting of why the ActionLibrary did not include financing decisions. The short answer is that modeling a realistic capital allocation strategy requires data that either is not publicly available at quarterly granularity or requires a financial domain model well beyond the scope of this project. Share repurchase programs, for example, involve board authorizations, market timing decisions, and treasury management practices that do not cleanly map to a single quarterly action with predictable state effects. Dividend management and debt issuance carry similar complexity. The decision was made to focus on operational and strategic actions where the inputs and effects were tractable, and to acknowledge the resulting structural gap explicitly rather than model it poorly. The conservative CFO ablation confirmed that this was a structural gap and not a behavioral one.

Margin compression was the second failure mode. The board prioritized growth which is directionally correct, but without the operational efficiency gains that real investment typically produces over time. Prompt B applies the immediate effects of actions but does not model the lagging productivity improvements that accompany sustained R&D and infrastructure investment. This is a modeling simplification, not an oversight.

6.3 The Restraint Gap

The most interesting finding is not about strategy. It is about restraint.

Real executives have quarterly guidance commitments that constrain how aggressively they can invest. They have board oversight that asks uncomfortable questions about cash burn. The simulation modeled none of these constraints. The board had the action space, the financial data, and the external context, but no accountability structure beyond the state it was accumulating.

The personality swap run provided a partial exception. The emergent self-correction in Q4 suggests that state deterioration can substitute for institutional guardrails, given sufficient severity. But it required four quarters of compounding to trigger. The guardrails in real organizations activate sooner.

6.4 On Role Identity vs Weight Vectors

The personality swap ablation was designed to test where behavioral differentiation actually lives in the pipeline. The answer is: everywhere, but in different ways.

The proposal phase is dominated by role identity. A CFO with CTO-style weights still writes like a CFO. The archetype description in the system prompt shapes language and framing in ways the weight vector cannot override.

The voting phase is dominated by weights. The swapped CFO applied less friction to R&D-intensive actions than the baseline CFO, consistent with its CTO-style weight vector.

7. Limitations

These are listed in order of how much they affected the results:

Cash modeling. The ActionLibrary omits all financing decisions. This was a scoping decision, not an oversight, and its consequences are fully visible in Tables 3-5. No amount of CFO conservatism can produce cash management behavior that is not in the action space.

Single-model simulation. All agents run on the same underlying model. Whether apparent role differentiation reflects genuine multi-agent dynamics or one model doing a committed impression of ten different executives is genuinely unclear. A future version using different models for different roles would be illuminating.

Revenue projection. Prompt B does not forecast revenue. The simulation compares strategic behavior and investment allocation, not financial outcomes. Treating the revenue figures as forecasts would be a category error.

Total employees. This field is not available in quarterly earnings files and was set manually from known annual figures. The simulation then compounds it forward, which is technically correct and also how companies end up with unexpectedly large headcounts.

Prompt B transient failures. Occasionally, Prompt B returns the input state unchanged due to malformed JSON output. Affected quarters show identical start and end values and must be re-run. This was resolved by retry in all final runs.

8. Conclusion

An AI C-suite can identify the right strategy. It cannot exercise the financial discipline required to execute it sustainably. With a more complete action space (one that includes financing decisions) the cash dimension would likely close substantially.

Across all simulation runs, the AI board correctly identified AI-heavy, cloud-forward investment as the appropriate strategic direction for FY2023 without access to any FY2023 data. Revenue grew in the correct direction every quarter. Strategic alignment with the actual company's priorities was the most consistent finding across all configurations.

What the board could not replicate was restraint. Cash was depleted in every run. Margins compressed across all configurations. Headcount grew more aggressively than actuals in most runs. These failures reflect both structural gaps in the action space (no financing decisions) and the absence of the implicit accountability mechanisms that constrain real executive behavior.

The personality swap ablation produced one genuinely surprising result: emergent self-correction. A board operating on a deteriorating cash position responded conservatively without external instruction, reversing headcount growth and recovering cash in Q4. This suggests that state pressure can partially substitute for institutional guardrails, given sufficient deterioration.

The code repo can be found here: https://github.com/oliviacarino/C-Suite

Future work should address the ActionLibrary's financing gap, test multi-model board configurations, and run multiple simulation trials to characterize output variance.

References

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